

SIH 2026 · TECHNICAL APPROACH

PAIMANA

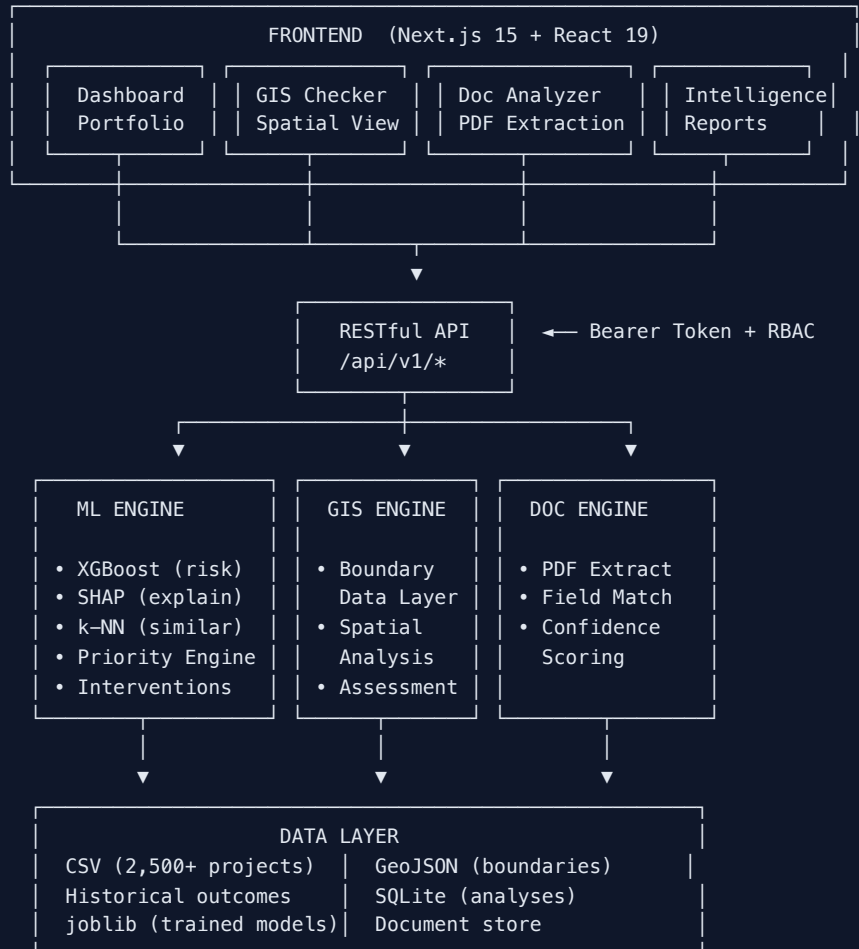
AI-Powered Project Intelligence for Indian Infrastructure Monitoring

Project Analytics & Intelligence for Monitoring Actions with National Accountability

Next.js · FastAPI · XGBoost · Shapely

September 2026

1 System Architecture



Feature 1 — AI Delay Risk Prediction

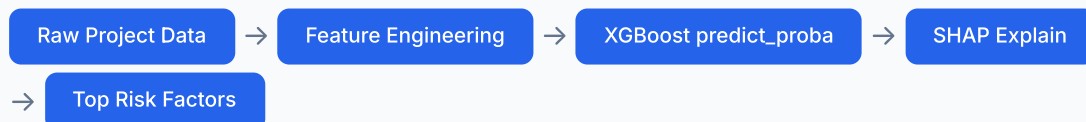
XGBoost

SHAP

scikit-learn

Aspect	Detail
Algorithm	XGBoost Binary Classifier (binary:logistic)
Input Features	land_acquisition_pct, milestone_slippage_count, financial_progress_gap_pct, pending_clearances_count, contractual_issues_count, months_elapsed
Output	Calibrated delay probability (0–1)
Explainability	SHAP TreeExplainer → top 5 risk factors with human-readable descriptions
Hyperparameters	120 estimators, max_depth=3, learning_rate=0.06, subsample=0.9
Training Data	2,500+ Indian infrastructure projects with labelled delay outcomes

Pipeline:

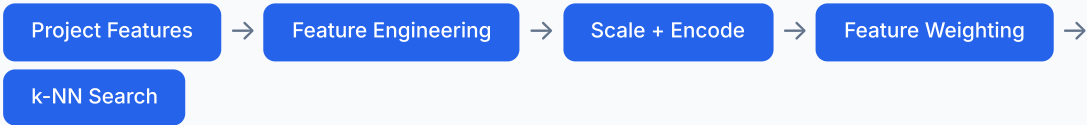


Feature 2 — Historical Similarity Engine

k-NN StandardScaler OneHotEncoder

Aspect	Detail
Algorithm	k-Nearest Neighbors (Euclidean distance, brute-force)
Preprocessing	StandardScaler (numeric) + OneHotEncoder (categorical: sector, state)
Engineered Features	cost_overrun_percentage, financial_physical_gap, milestone_delay_ratio, project_age_ratio
Feature Weighting	Domain-tuned: milestone_delay_ratio (2.5×), land_acquisition_pending (2.0×), funding_issue (2.0×)
Output	Top-k similar historical projects + aggregated evidence statistics

Pipeline:



Feature 3 — Priority & Decision Engine

Deterministic Weighted Scoring

Fully transparent, no ML — combines evidence from Features 1, 2, and 5 into one composite score.

Component	Weight	Source
AI Delay Risk	0.35	XGBoost probability × 100
Predicted Delay Severity	0.20	P(delay) × avg_historical_delay
Cost Overrun Risk	0.15	Historical evidence
Milestone Health	0.15	delayed / total milestones
GIS Spatial Risk	0.15	Environmental collision severity

Output: Composite score (0–100) → Critical High Medium Low

Feature 4 — GIS Boundary Data Layer

Shapely GeoJSON PostGIS

Aspect	Detail
Storage	Dual-backend: GeoJSON file store (default) / PostGIS (production)
Geometry Model	Polygon + MultiPolygon with full RFC 7946 validation
Validation	Ring closure, winding order normalization, self-intersection rejection, coordinate bounds checking
Categories	Wildlife sanctuary, national park, tiger reserve, forest reserve, wetland, heritage site, coastal zone, tribal area, defence zone, river/floodplain
Spatial Index	Shapely STRtree / PostGIS GiST (bbox pre-filter → exact geometry test)
ID Strategy	UUIDv5 (deterministic from source + name + category)

Feature 5 — Spatial Analysis Engine

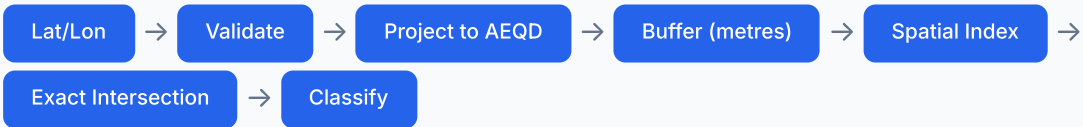
Shapely pyproj AEQD Projection

Aspect	Detail
CRS Handling	WGS84 → AEQD (centred on project point) → metres → back to WGS84
Key Principle	<i>"Metres are never applied to degrees"</i> — buffer only in projected CRS
Index Strategy	Bounding-box pre-filter (over-inclusive) → exact geometry intersection

Collision Classification:

Type	Description
DIRECT COLLISION	Project point lies inside a protected boundary
BUFFER COLLISION	Project buffer zone intersects a boundary
PROXIMITY WARNING	Boundary within proximity threshold
CLEAR	No nearby boundaries detected

Pipeline:



Feature 6 — Intervention Recommendations

Rule-Based Deterministic No LLM

Evidence-linked, ranked actions. Statutory obligations above managerial ones.

Rule	Trigger Condition
Spatial Clearance	GIS boundary screening requires clearance
Site Review	Direct collision with protected area
Land Acquisition	<code>land_acquisition_pending = true</code>
Regulatory Clearance	<code>clearance_pending = true</code>
Milestone Recovery	Milestone delay ratio $\geq$ threshold
Funding Release	<code>funding_issue = true</code>
Cost Baseline Review	Cost overrun $\geq$ threshold
Contractor Review	<code>contractor_issue = true</code>
Schedule Review	Late schedule + high risk probability

Feature 7 — Document Intelligence (PDF Extraction)

pdfplumber Label Matching

Aspect	Detail
Method	Label-based matching over text lines and table rows — no guessing
Extracted Fields	All ProjectRiskRequest fields (costs, durations, milestones, coordinates)
Confidence	Each field tagged high / low + explanatory note
Unit Handling	Auto-conversion: crore $\leftrightarrow$ lakh $\leftrightarrow$ million $\leftrightarrow$ billion
Human Review	Extracted values presented for confirmation before reaching ML pipeline

Feature 8 — SLA Monitoring & Notifications

MSG91 SMS Threshold Rules

Aspect	Detail
SLA Engine	Deadline tracking per project milestone
Notification	MSG91 SMS API (configurable)
Amber Threshold	30 days remaining before deadline
Critical Threshold	Risk probability $\geq$ 80%

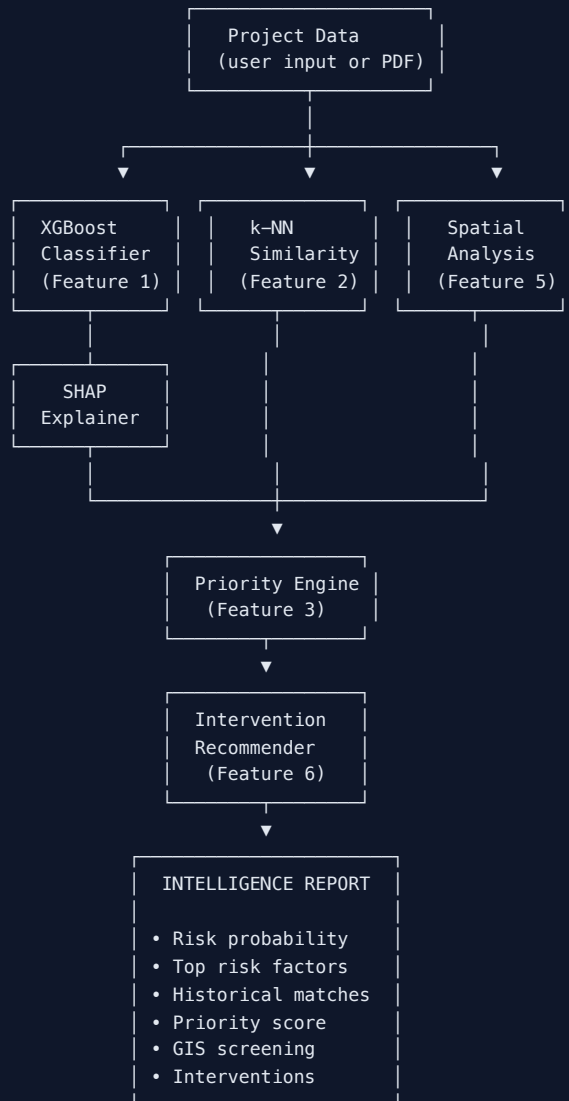
3 Technology Stack

Layer	Technology
Frontend	Next.js 15, React 19, TypeScript, TailwindCSS v4, Lucide Icons
Backend	Python, FastAPI, Uvicorn (ASGI server)
ML / AI	XGBoost, scikit-learn, SHAP, NumPy, Pandas
Geospatial	Shapely, pyproj, GeoPandas (optional), PostGIS (optional)
Document Processing	pdfplumber (text + table extraction from PDFs)
Data Storage	CSV (projects), GeoJSON (boundaries), SQLite (analyses), joblib (models)
Authentication	Bearer Token + RBAC (SHA-256 hashed registry)
Notifications	MSG91 SMS API
Validation	Pydantic v2 (strict schema enforcement)

4 API Endpoints

Endpoint Group	Purpose
/api/v1/predict	Delay risk prediction
/api/v1/similar-projects	Historical similarity search
/api/v1/project-intelligence	Combined intelligence report (all engines)
/api/v1/priority	Priority scoring & ranking
/api/v1/gis/*	Boundary CRUD, spatial analysis, collision checks
/api/v1/documents/*	Document upload, extraction, repository
/api/v1/analyses/*	Saved analyses management
/api/v1/projects	Project catalogue
/api/v1/sla/*	SLA status and monitoring
/health	System health check

5 Intelligence Report — Data Flow



6 Security Architecture

- **Bearer Token Authentication** — SHA-256 hashed token registry, loaded once at startup
- **Role-Based Access Control (RBAC)** — Roles: admin, analyst, viewer
- **Permission-gated endpoints** — Routes declare required permissions, not raw token checks
- **CORS policy** — Restricted to application origins only
- **Designed as a seam** — Swappable to OIDC/JWT without modifying any route handler

7 Key Design Principles

1

Deterministic & Auditable

Identical inputs → identical outputs. Every threshold lives in a config file. No randomness in production inference.

2

Explainability First

SHAP for ML predictions; template-based rationale for interventions; geometry-derived reasons for GIS flags.

3

Separation of Concerns

ML models → services → API routes → frontend. Each layer has a single responsibility.

4

Graceful Degradation

If any service (ML, GIS, similarity) fails to load, the app starts in degraded mode and reports exactly what's unavailable.

5

No Silent Repairs

GIS geometry is rejected, not silently fixed. Document extraction reports low confidence, not a guess.

6

Domain-Weighted Features

Similarity engine weights features by their empirically validated association with delay outcomes in historical data.

8

Frontend Pages

Page	Description
Login	Secure workspace access with email/password
Dashboard	Portfolio view of all projects with search, sector/state filters
Risk Signals	Filtered view showing only projects with active risk indicators
Project Intelligence	Full per-project report: risk, SHAP factors, matches, priority, interventions, GIS
Geospatial Checker	Interactive GIS collision analysis with boundary visualization
Document Analyzer	PDF upload → auto extraction → human review → feed to prediction
Scenario Lab	What-if analysis for project parameters

Settings	Workspace and user configuration
----------	----------------------------------